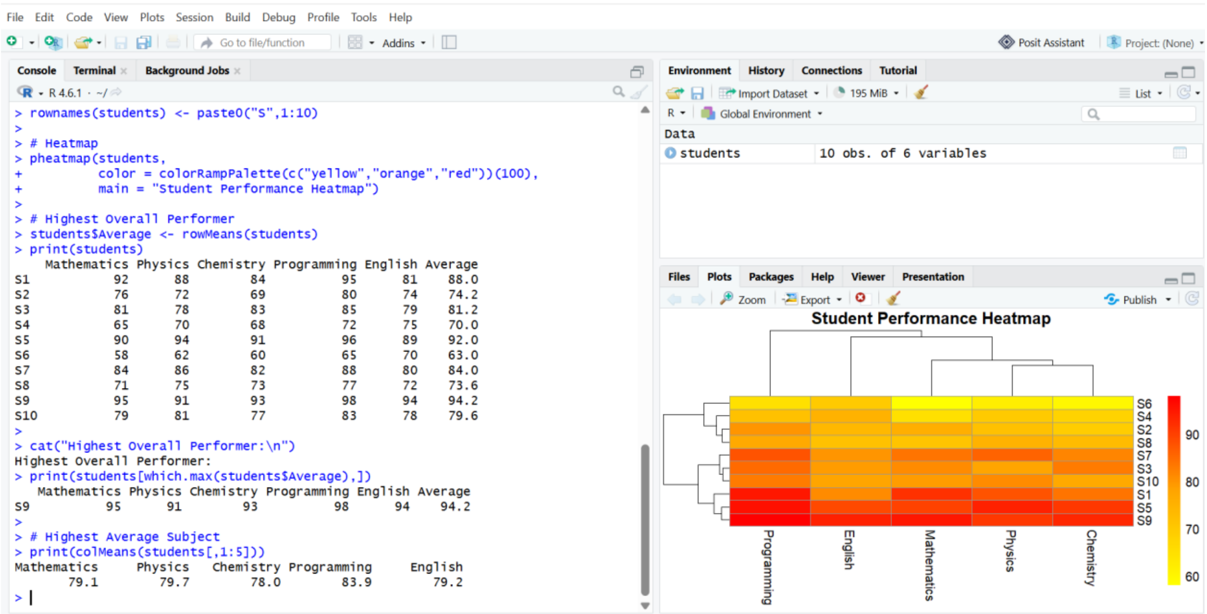
1. Heatmap – Student Performance Analysis

Code:

```
install.packages("pheatmap")

library(pheatmap)

students <- data.frame(
  Mathematics = c(92,76,81,65,90,58,84,71,95,79),
  Physics = c(88,72,78,70,94,62,86,75,91,81),
  Chemistry = c(84,69,83,68,91,60,82,73,93,77),
  Programming = c(95,80,85,72,96,65,88,77,98,83),
  English = c(81,74,79,75,89,70,80,72,94,78))

rownames(students) <- paste0("S",1:10)

pheatmap(students,
         color = colorRampPalette(c("yellow","orange","red"))(100),
         main = "Student Performance Heatmap")

students$Average <- rowMeans(students)

print(students)

cat("Highest Overall Performer:\n")

print(students[which.max(students$Average),])

print(colMeans(students[,1:5]))
```



1. Construct a Heatmap

- A heatmap represents data using different colors.
- Higher marks are shown with darker colors.
- Lower marks are shown with lighter colors.
- It helps compare students and subjects easily.

2. Highest Overall Performer

- Student **S9** has the highest average marks.
- **Average = 94.2 marks**

3. Highest Average Subject

- **Programming** has the highest average marks.
- **Average = 83.9 marks**

4. Students Needing Support

- **S6** (Average = 63.0)
- **S4** (Average = 70.0)

5. Advantages of Heatmap

- Easy to compare multiple values.
- Highlights high and low scores quickly.
- Identifies trends and patterns.
- Suitable for large datasets.

2. Violin Plot & Boxplot – Employee Salary Analysis

Code:

```
install.packages("ggplot2")
install.packages("reshape2")
library(ggplot2)
library(reshape2)
salary <- data.frame(
  HR = c(4.2,4.5,4.7,4.1,4.6,4.3,4.8,4.4,4.9,4.5),
  IT = c(6.5,7.0,7.4,6.8,8.2,7.6,8.5,7.1,8.8,7.3),
  Finance = c(5.8,6.2,6.0,5.9,6.7,6.4,6.9,6.1,7.2,6.3),
```

Marketing = c(5.1,5.3,5.5,5.4,5.8,5.6,5.9,5.2,6.0,5.4))

salary_long <- melt(salary)

ggplot(salary_long, aes(variable, value, fill=variable)) +

geom_violin() +

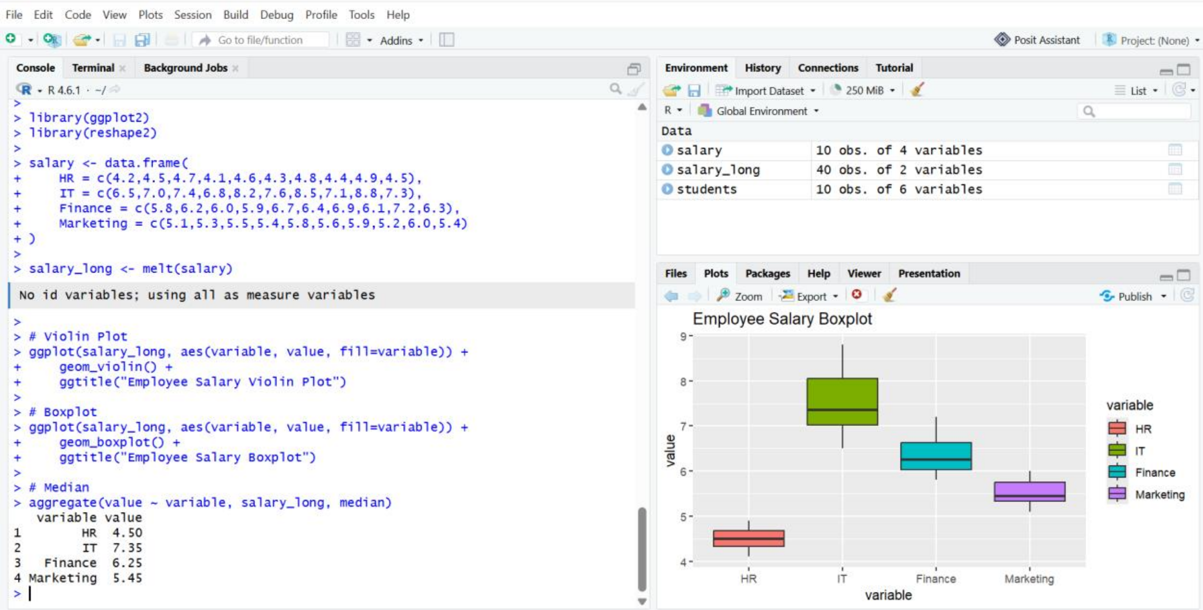
ggtitle("Employee Salary Violin Plot")

ggplot(salary_long, aes(variable, value, fill=variable)) +

geom_boxplot() +

ggtitle("Employee Salary Boxplot")

aggregate(value ~ variable, salary_long, median)



1. Violin Plot

- Shows the distribution of salary data.
- Displays the density of values.
- Helps compare salary distributions across departments.

2. Boxplot

- Shows the median.
- Displays quartiles and spread.
- Identifies possible outliers.

3. Largest Variation

- **IT Department** has the largest salary variation.

4. Highest Median

- **IT Department**
- Median salary ≈ **7.35 Lakhs**

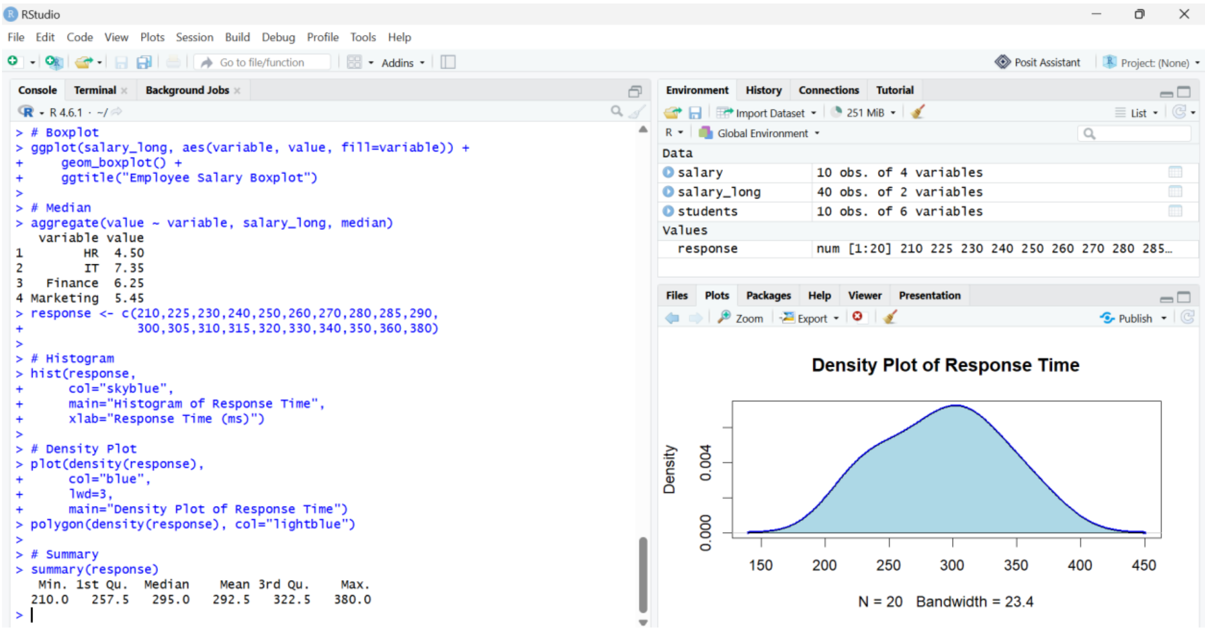
5. Better Plot for Distribution

- **Violin Plot**
- Shows distribution shape and density more clearly than a boxplot.

3. Histogram & Density Plot – Website Response Time

Code:

```
response <-
c(210,225,230,240,250,260,270,280,285,290,300,305,310,315,320,330,340,350,360,380)

hist(response,
    col="skyblue",
    main="Histogram of Response Time",
    xlab="Response Time (ms)")

plot(density(response),
    col="blue",
    lwd=3,
    main="Density Plot of Response Time")

polygon(density(response), col="lightblue")

summary(response)
```



1. Histogram

- Displays the frequency of response times.
- Groups data into intervals (bins).

2. Density Plot

- Shows a smooth distribution curve.
- Helps identify where most values are concentrated.

3. Skewness

- Slightly **positively (right) skewed**.

4. Common Range

- Most response times are between **250 ms and 320 ms**.

5. Better Plot for Distribution

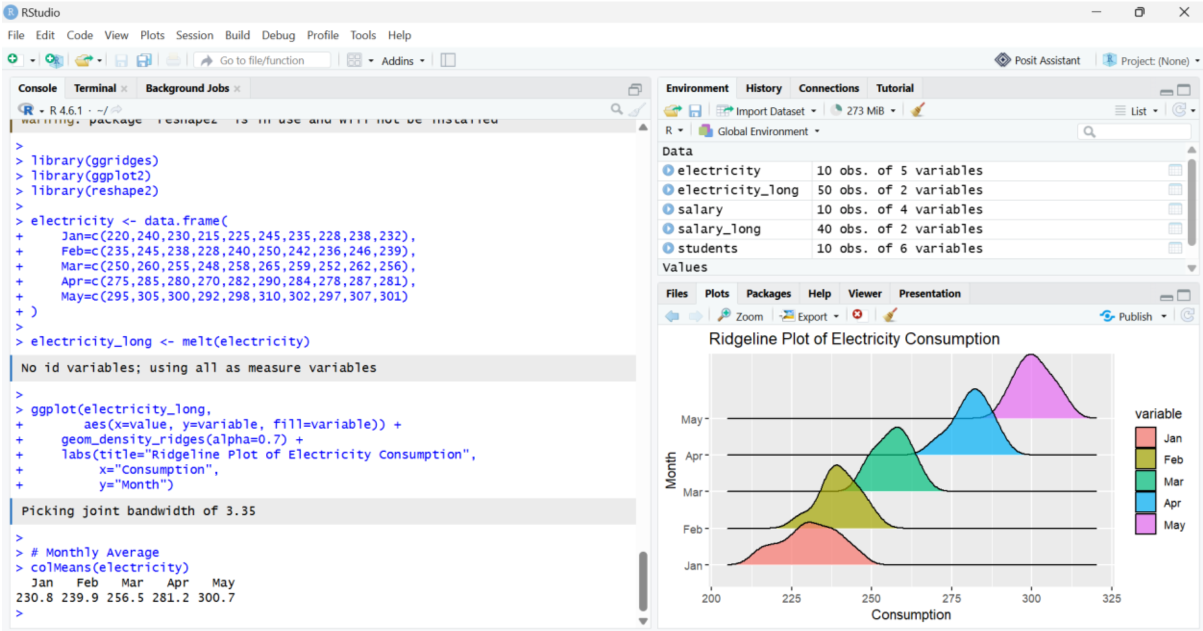
- **Density Plot**
- Provides a smoother and clearer view of the overall distribution.

4. Ridgeline Plot – Electricity Consumption

Code:

```
install.packages("ggridges")
install.packages("reshape2")
library(ggridges)
library(ggplot2)
library(reshape2)
electricity <- data.frame(
  Jan=c(220,240,230,215,225,245,235,228,238,232),
  Feb=c(235,245,238,228,240,250,242,236,246,239),
  Mar=c(250,260,255,248,258,265,259,252,262,256),
  Apr=c(275,285,280,270,282,290,284,278,287,281),
  May=c(295,305,300,292,298,310,302,297,307,301))
electricity_long <- melt(electricity)
ggplot(electricity_long,
       aes(x=value, y=variable, fill=variable)) +
```

```
  geom_density_ridges(alpha=0.7) +
  labs(title="Ridgeline Plot of Electricity Consumption",
       x="Consumption",
       y="Month")
colMeans(electricity)
```



1. Ridgeline Plot

- Compares monthly electricity consumption distributions.
- Displays overlapping density curves for each month.

2. Highest Average Month

- **May**
- Average consumption = **300.7 units**

3. Widest Spread

- **January**
- Range = **30 units**

4. Trend

- Electricity consumption increases steadily from **January to May**.
- Indicates increasing usage over the months.

5. Why Ridgeline Plot?

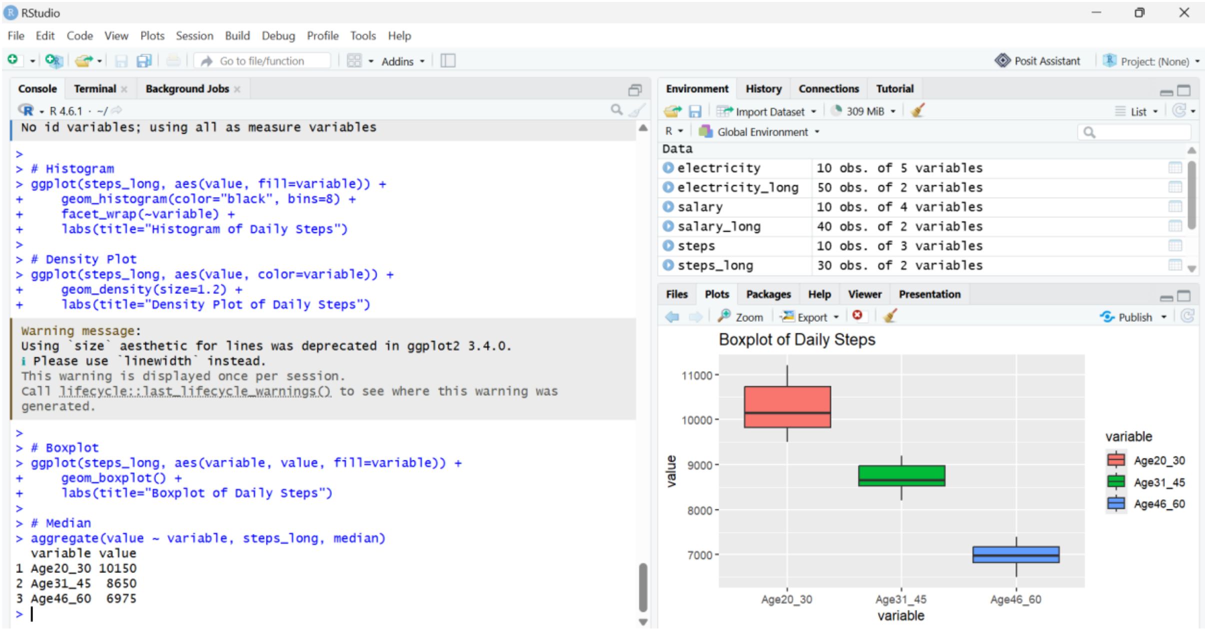
- Compares multiple distributions at once.
- Shows monthly changes clearly.

- Makes it easy to identify trends and patterns.

5. Histogram, Density & Boxplot – Daily Steps

Code:

```
install.packages("reshape2")
install.packages("ggplot2")
library(reshape2)
library(ggplot2)
steps <- data.frame(
  Age20_30=c(9500,10200,9800,11000,10500,9700,10100,10800,11200,9900),
  Age31_45=c(8200,8600,8400,9000,8900,8500,8700,9100,9200,8600),
  Age46_60=c(6500,6700,6900,7200,7000,6800,7100,7300,7400,6950))
steps_long <- melt(steps)
ggplot(steps_long, aes(value, fill=variable)) +
  geom_histogram(color="black", bins=8) +
  facet_wrap(~variable) +
  labs(title="Histogram of Daily Steps")
ggplot(steps_long, aes(value, color=variable)) +
  geom_density(size=1.2) +
  labs(title="Density Plot of Daily Steps")
ggplot(steps_long, aes(variable, value, fill=variable)) +
  geom_boxplot() +
  labs(title="Boxplot of Daily Steps")
aggregate(value ~ variable, steps_long, median)
```



1. Histograms

- Show the frequency distribution of daily steps.
- Useful for comparing activity levels across age groups.

2. Density Curves

- Show a smooth distribution of daily steps.
- Help identify where step counts are concentrated.

3. Boxplots

- Show median values.
- Display spread and variability.
- Help detect outliers.

4. Highest Median Group

- **20–30 years**
- Median = **10,150 steps**

5. Best Visualization

- **Boxplot**